3 The needle-carrier of a sewing machine is constrained to move in a vertical line by fixed guides as shown in Figure 3.1. The simple harmonic motion of the needle-carrier is produced by a peg attached to the rim of a rotating disc which moves in a circle and engages the slot of the needle-carrier.

Figure 3.2 shows the front view of the positions of the peg and needle-carrier at intervals of a quarter period.

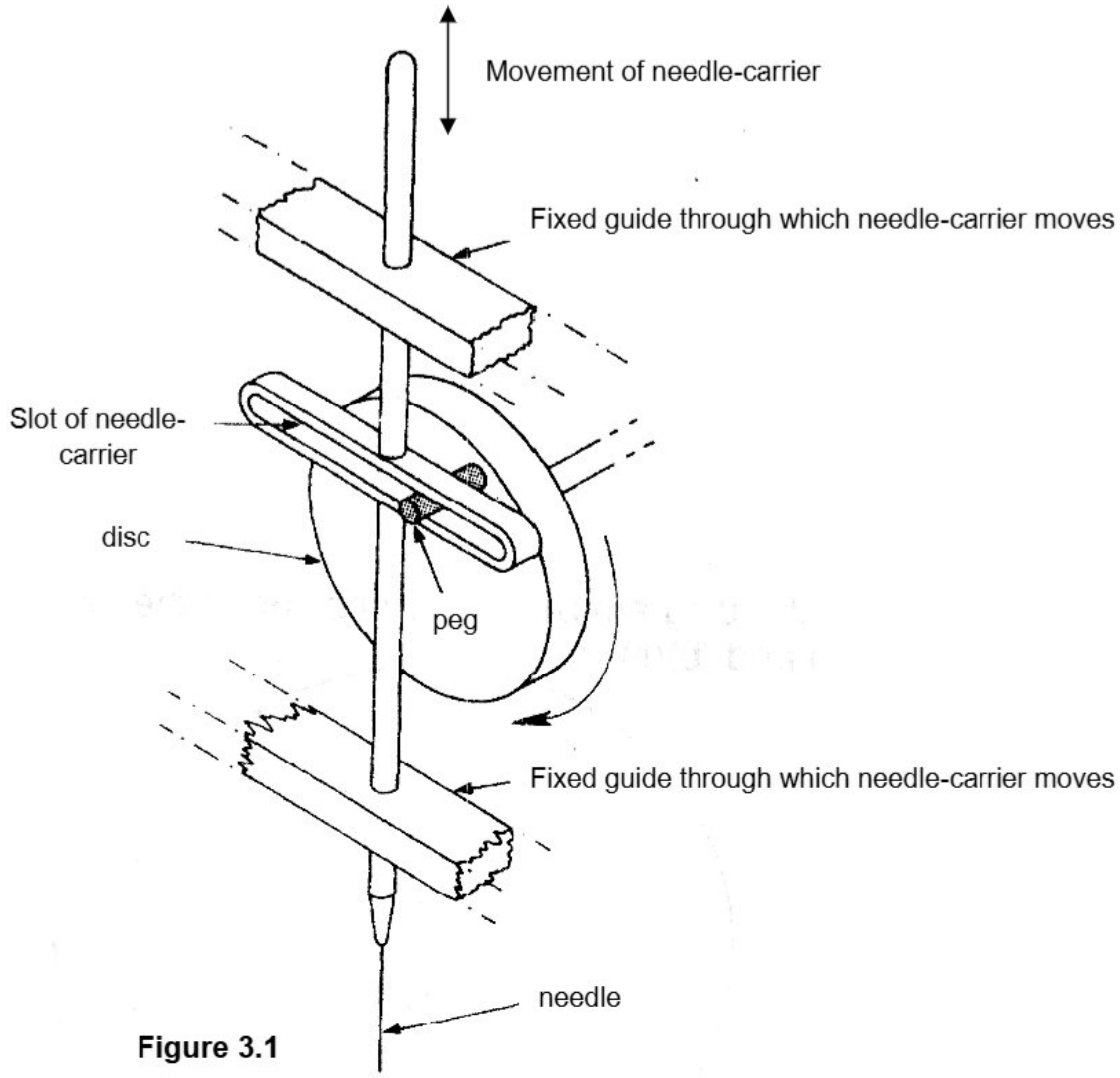



**Figure 3.1**

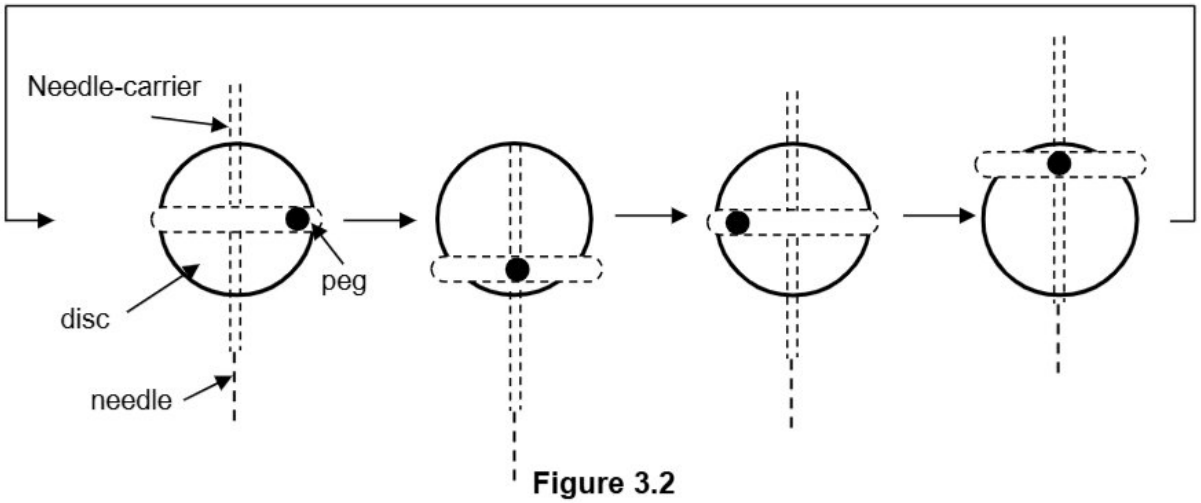



**Figure 3.2**

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**(a)** The machine makes 12 stitches per second.

Show that the angular speed of the peg's circular motion is 75 rad $s^{-1}$.

[1]

**(b)** From its highest to lowest point, the needle-carrier moves through a vertical distance of 32 mm.

**(i)** Calculate the maximum acceleration of the needle-carrier.

maximum acceleration = ..... m $s^{-2}$ [2]

**(ii)** Explain why the peg exerts a larger contact force on the slot of the needle-carrier when it is at its lowest position compared to when it is at its highest position.

.....

.....

.....

..... [2]

**(c)** The needle-carrier and needle have a total mass of 25 g.

Sketch in Figure 3.3 the variation of the kinetic energy of the needle-carrier and needle with time $t$ for two complete periods. The needle-carrier and needle are at the highest position at $t = 0$.

Indicate all relevant values on the axes.

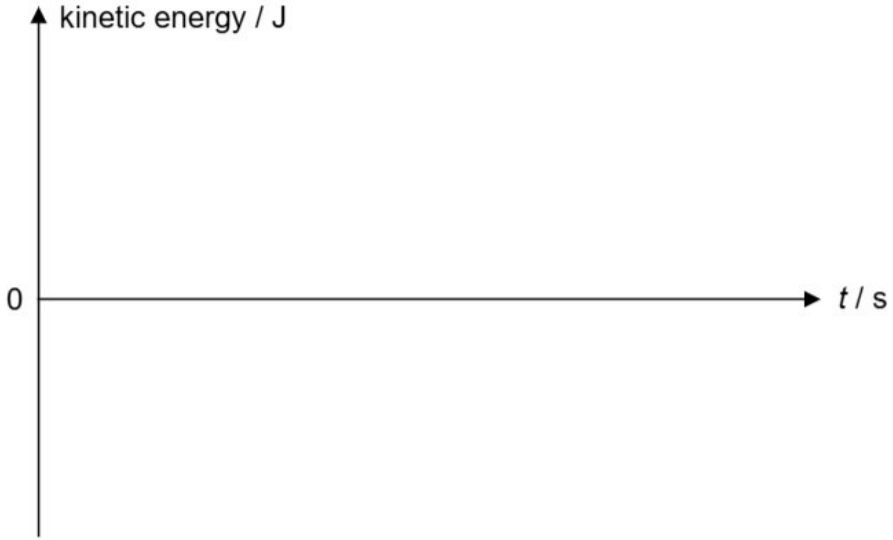



**Figure 3.3**

[3]

[Total: 8]

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